

7 (a) (i) State what is meant by isotopes.

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.....

..... [2]

(ii) A thorium-232 nucleus ($^{232}_{90}\text{Th}$) undergoes radioactive decay to form another nucleus. A sequence of further decays occurs until a nucleus is formed that is another isotope of thorium. In each decay, either a single α -particle or a single β^- particle is emitted.

Determine the minimum numbers of α -particles and β^- particles that could be emitted during the sequence of decays to form the other isotope of thorium from thorium-232.

minimum number of α -particles =

minimum number of β^- particles = [2]

(iii) Complete Figure 7.1 to show the nucleon number and proton number of the thorium isotope formed by the sequence of decays.

..... Th
.....

Figure 7.1

[2]

(b) (i) State the quark composition of a proton.

..... [1]

(ii) Describe the change to the quark composition of the proton when it undergoes β^+ decay.

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..... [1]

[Total: 8]